Examiner's Use

5 **(a)** State the relationship between the energy E and momentum p of a photon.

.....[1]

(b) An electron in a quantum dot is modelled as a particle in a one-dimensional infinite square well of width 3.0 nm.

It makes a transition from quantum state $n = 8$ to $n = 2$ and emits a photon.

An array of identical quantum dots produces many such photons, which are incident on a metal plate of work function 2.30 eV.

(i) Calculate the threshold frequency of the metal.

threshold frequency = ... Hz [2]

(ii) Show that the energy of the photon is 2.51 eV.

[3]

(iii) Hence, determine the de Broglie wavelength of the fastest photoelectrons.

de Broglie wavelength = ... m [3]

(iv) The photoelectrons are sent through a velocity selector before being directed at a double-slit forming an interference pattern on a screen.

State and explain the effect on the interference pattern when the velocity selector is removed.

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..... [3]

[Total: 12]